2026 Senate forecast - Most_Magnificent_Model

scheduled

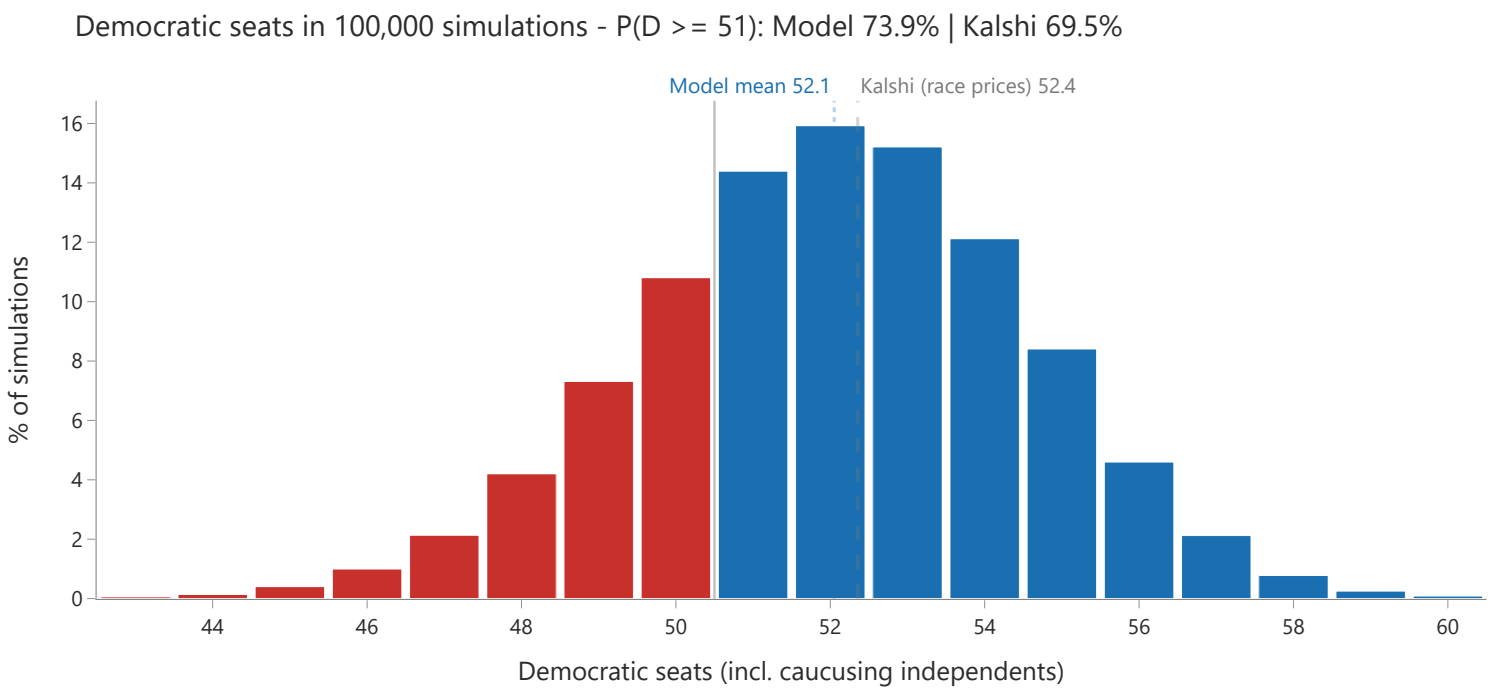
Team 5: Elizabeth Miller, Mark Scott, Ethan Verrone, Boaz Alkalay, Leo Westebbe

scheduled run, 2026-10-30 11:10

P(DEMOCRATS >= 51 SEATS)	EXPECTED DEMOCRATIC SEATS	KALSHI CONTROL MARKET	RUN
73.9%	**52.1**	**69.5%**	**2026-10-30 11:10**
simulation precision ±0.3 pts	median 52 | 80%: 49-55 | 95%: 47-57	sum of race prices: 52.4 seats	100,000 simulations | data as of 2026-10-30 | 5 min

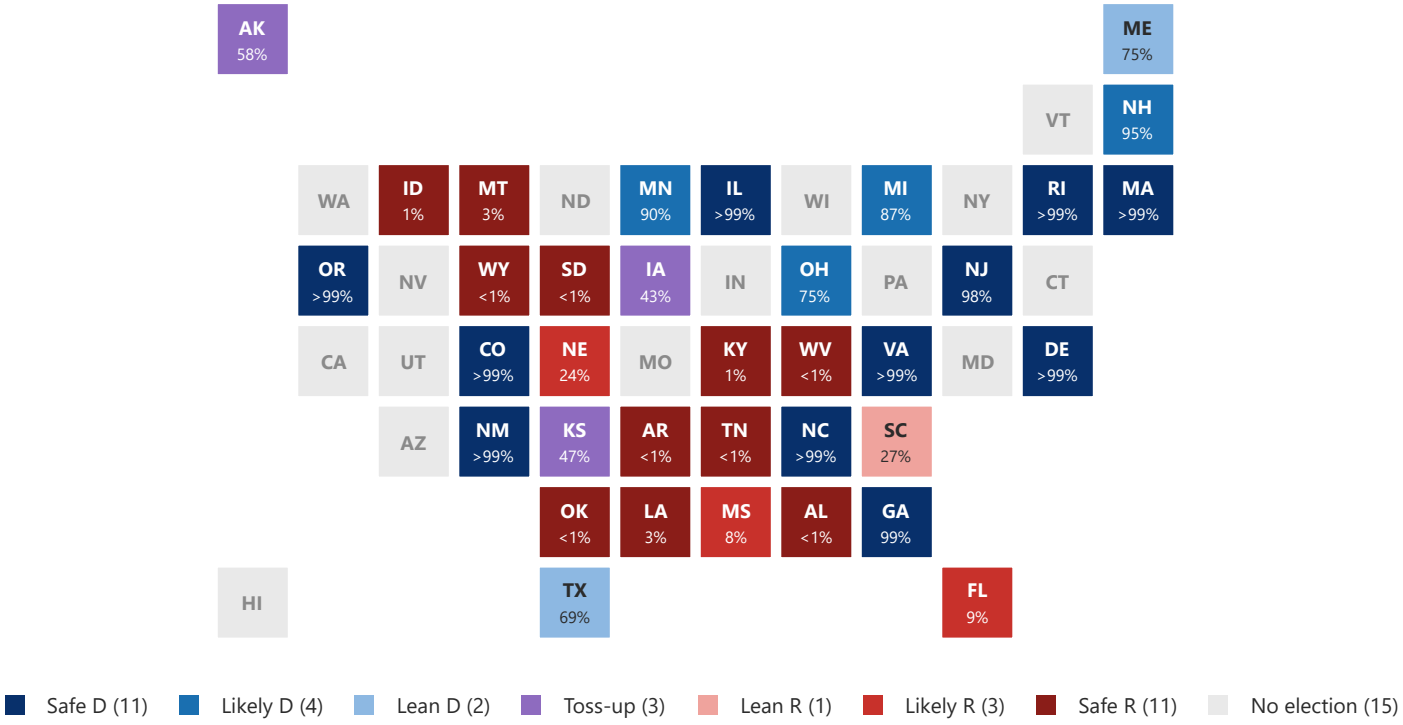
The model and Kalshi expect about the same number of Democratic seats (52.1 vs 52.4); the model's higher control odds come from rating Michigan, Maine and Ohio higher for Democrats than the market does.

Seat distribution

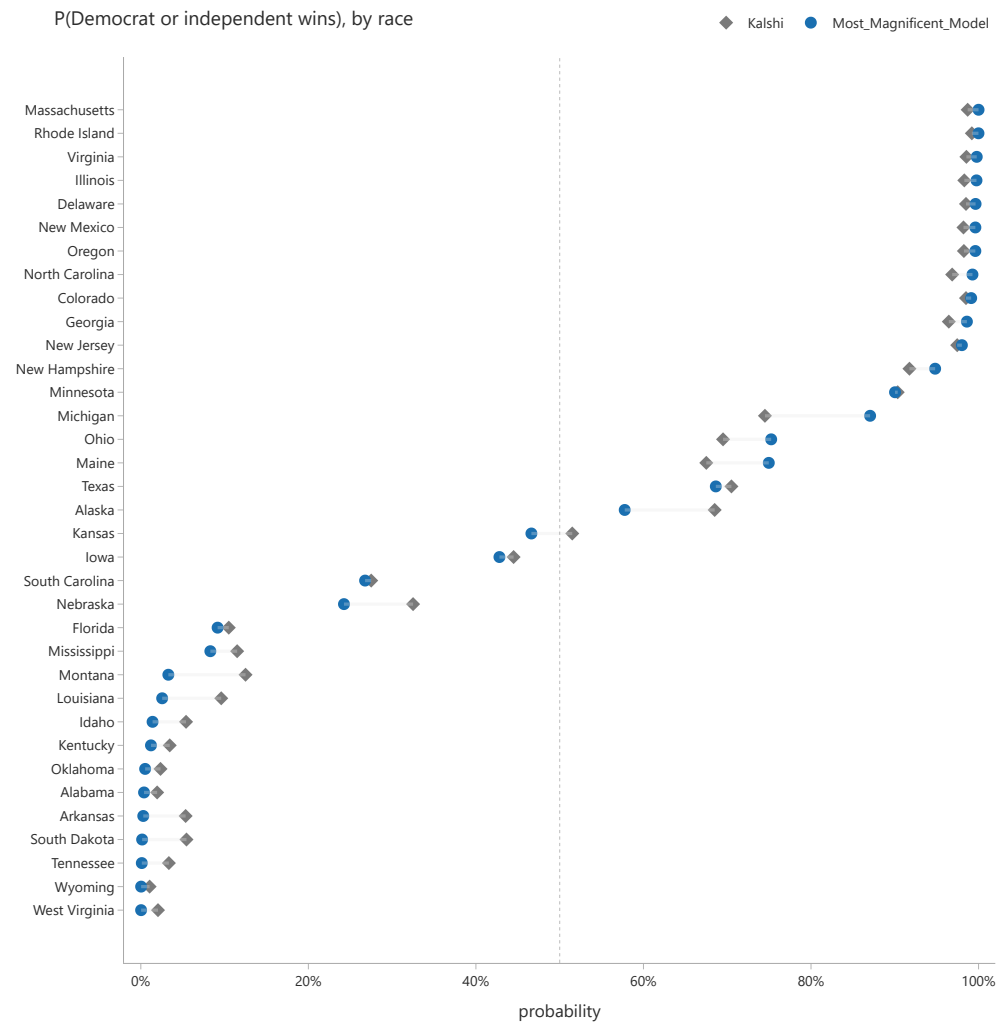


Map

Race ratings (number = Model P(D/Ind wins); D includes independents)
Toss-up under 60%, Lean 60%-75%, Likely 75%-95%, Safe 95%+



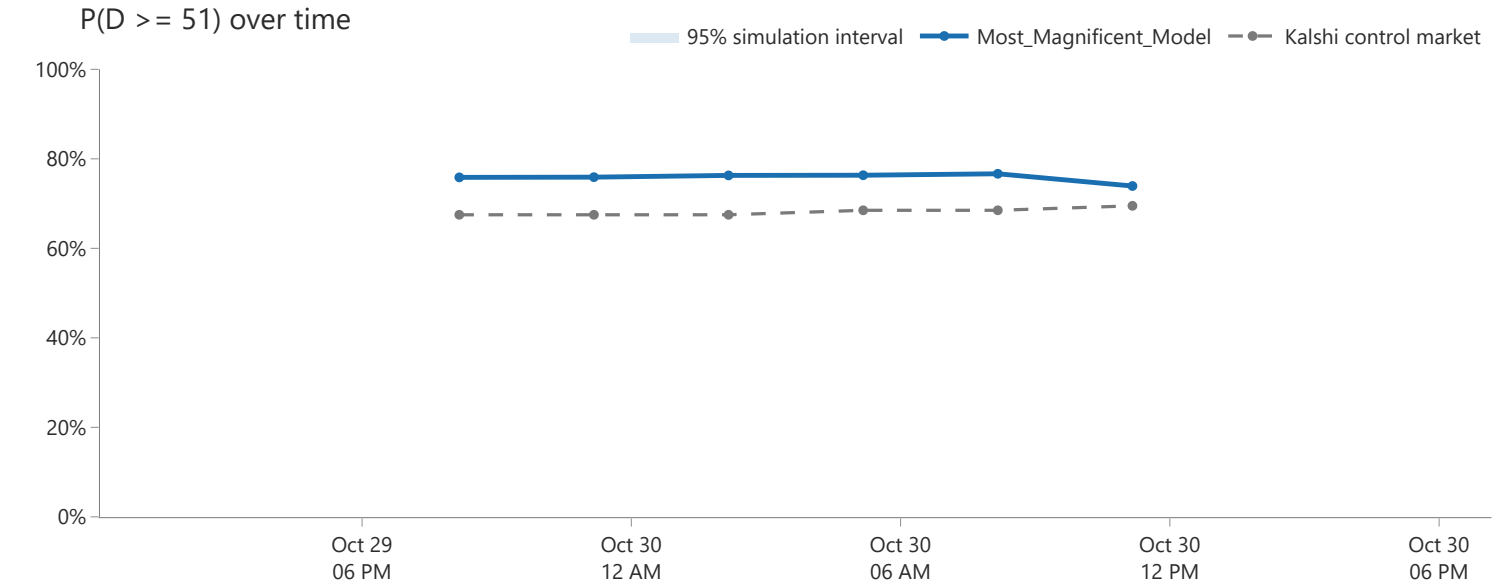
Race by race

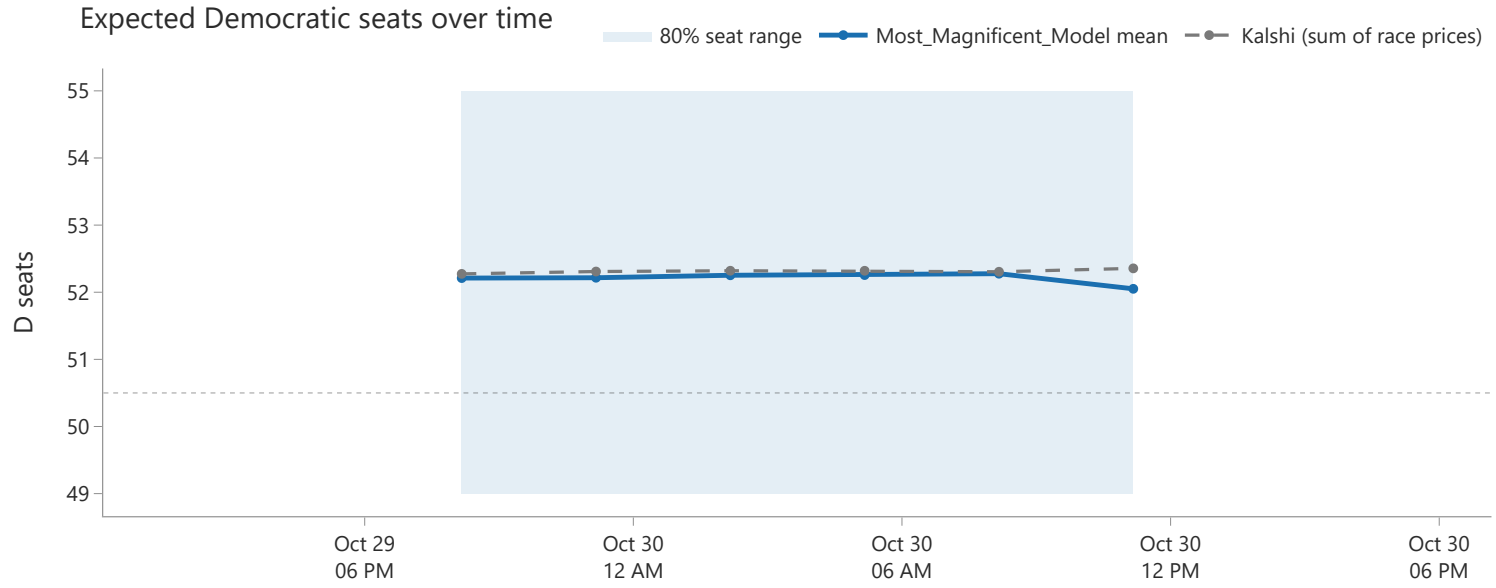


Race	Call	Model P(D/Ind)	Kalshi P(D/Ind)	Model - Kalshi (pts)	Projected margin	+/- (1 SD)	Polls	Model change since first run (pts)	Kalshi change since first run (pts)
Massachusetts	Safe D	>99%	99%	+1	D+34.0	+/-5.7	13	+0	+0
Rhode Island	Safe D	>99%	>99%	+1	D+25.1	+/-5.5	4	+0	+0
Virginia	Safe D	>99%	99%	+1	D+18.5	+/-5.9	1	-0	+0
Illinois	Safe D	>99%	98%	+1	D+17.2	+/-5.7	2	-0	+0
Delaware	Safe D	>99%	98%	+1	D+27.8	+/-9.4	0	-0	+0
New Mexico	Safe D	>99%	98%	+1	D+18.5	+/-6.4	2	-0	-0
Oregon	Safe D	>99%	98%	+1	D+27.8	+/-9.4	0	-0	-1

Race	Call	Model P(D/Ind)	Kalshi P(D/Ind)	Model - Kalshi (pts)	Projected margin	+/- (1 SD)	Polls	Model change since first run (pts)	Kalshi change since first run (pts)
North Carolina	Safe D	>99%	97%	+2	D+12.1	+/-4.7	40	-0	+0
Colorado	Safe D	>99%	98%	+1	D+23.8	+/-9.4	0	-0	+0
Georgia	Safe D	99%	96%	+2	D+10.8	+/-4.8	18	-0	+0
New Jersey	Safe D	98%	97%	+1	D+19.8	+/-9.4	0	-0	-0
New Hampshire	Likely D	95%	92%	+3	D+8.2	+/-5.1	21	-0	+1
Minnesota	Likely D	90%	90%	-0	D+6.2	+/-4.9	7	-0	+1
Michigan	Likely D	87%	74%	+13	D+5.2	+/-4.7	42	-1	+0
Ohio	Likely D	75%	70%	+6	D+3.1	+/-4.6	32	-1	+0
Maine	Lean D	75%	68%	+7	D+3.0	+/-4.6	18	-0	+4
Texas	Lean D	69%	70%	-2	D+2.2	+/-4.5	51	-1	+2
Alaska	Toss-up	58%	68%	-11	D+0.9	+/-4.7	18	-2	+0
Kansas	Toss-up	47%	52%	-5	R+0.4	+/-5.0	5	-1	+3
Iowa	Toss-up	43%	44%	-2	R+0.8	+/-4.7	24	-3	-3
South Carolina	Lean R	27%	28%	-1	R+2.8	+/-4.7	4	-2	+0
Nebraska	Likely R	24%	32%	-8	R+3.4	+/-5.1	4	-1	+0
Florida	Likely R	9%	10%	-1	R+6.1	+/-4.6	17	-1	+0
Mississippi	Likely R	8%	11%	-3	R+7.4	+/-5.5	3	-1	+0
Montana	Safe R	3%	12%	-9	R+10.1	+/-5.5	25	-0	+0
Louisiana	Safe R	3%	10%	-7	R+13.9	+/-7.1	0	-0	+0
Idaho	Safe R	1%	5%	-4	R+15.7	+/-6.9	1	-0	-0
Kentucky	Safe R	1%	3%	-2	R+22.0	+/-9.4	0	-0	+0
Oklahoma	Safe R	<1%	2%	-2	R+26.0	+/-9.4	0	-0	-0
Alabama	Safe R	<1%	2%	-2	R+16.5	+/-5.8	5	-0	-0
Arkansas	Safe R	<1%	5%	-5	R+14.7	+/-5.0	5	-0	+1
South Dakota	Safe R	<1%	5%	-5	R+16.2	+/-5.1	3	-0	-0
Tennessee	Safe R	<1%	3%	-3	R+23.3	+/-6.9	1	-0	-0
Wyoming	Safe R	<1%	1%	-1	R+38.0	+/-9.4	0	-0	-0
West Virginia	Safe R	<1%	2%	-2	R+37.8	+/-9.4	0	-0	+0

History


P(D >= 51) over time
95% simulation interval
Most_Magnificent_Model
Kalshi control market
100%
80%
60%
40%
20%
0%
Oct 29
06 PM
Oct 30
12 AM
Oct 30
06 AM
Oct 30
12 PM
Oct 30
06 PM


Expected Democratic seats over time
80% seat range
Most_Magnificent_Model mean
Kalshi (sum of race prices)
D seats
55
54
53
52
51
50
49
Oct 29
06 PM
Oct 30
12 AM
Oct 30
06 AM
Oct 30
12 PM
Oct 30
06 PM

Notes

- Model Testing: 34 PASS | 7 WARN | 0 FAIL (details in model_warnings.xlsx).
- Data: votehub_senate live, votehub_generic_ballot live, votehub_approval live, kalshi_races live, kalshi_national live.

Key settings

Poll source	VoteHub API
Poll age	all polls, half-life 7 days
Polls : fundamentals : Kalshi	Bayesian precision (poll floor 3.9, prior SD 9.3); Kalshi precision 1/8^2, longshot-corrected, conversion SD >= 4.42
Poll reliability (thin races)	automatic (Bayesian precision)
Fundamentals	PVI x 2 + national environment (generic-ballot blend 0.75) + candidate effects x 0.30
Incumbency	3.8 pts (appointed 1)
LV : RV : A	1 : 0.85 : 0.7; RV shift +0.9 (measured)
Flooding penalty	1/k^0.5
House effects	in-cycle, shrunk to scorecard (k 5)
Ungraded pollsters	random
National signals	VoteHub polls + Kalshi averages (= 5 polls)
Kalshi races	Bayesian input + comparison (found automatically)
Source errors	correlated (rho 0.3); sims: national 2.5 + region 1.5 + state, in pts
Calibration	calibration.json (2026-10-29 22:45)
Simulations	100,000

Equations and weights

1. WEIGHT OF EACH POLL

$$w = 0.5^{\text{age}/7} \times \left(\frac{\min(n,\,2500)}{800}\right)^{0.5} \times w_{\text{pop}} \times w_{\text{partisan}} \times w_{\text{generic}} \times w_{\text{field}} \times w_{\text{quality}} \times w_{\text{accuracy}} \times \frac{1}{k^{0.5}} \times \frac{1}{\text{versions}}$$

Population: LV 1, RV 0.85, adults 0.7. Partisan sponsor 0.6. Generic-party question 0.5. Full field 0.9. Quality 0.5 to 1 by VoteHub grade (ungraded firms get a random graded firm's grade in each simulation). Accuracy = (1.25 x 4 / poll SD)^2, kept between 0.5 and 2. k = the pollster's polls of that race.

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2. EACH POLL'S MARGIN

$$m = (D - R)\,\frac{D + R + U}{D + R} + s_{RV} \pm s_{\text{partisan}} - h, \qquad h = \frac{\sum \text{residuals} + 5\,h_{\text{scorecard}}}{n_{\text{polls}} + 5}$$

U = undecided. RV shift s_RV = +0.88 pts (registered-voter polls only; measured from LV/RV pairs). Partisan shift = 3 pts against the sponsor. Residual = poll minus the same state's average of other pollsters, from the last 60 days.

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3. POLLING AVERAGE

$$\bar{m} = \frac{\sum w\,m}{\sum w}, \qquad SD_{\text{polls}} = \sqrt{\frac{SD_{\text{poll}}^2}{n_{\text{eff}}} + 3.87^2 + SD_U^2}, \qquad n_{\text{eff}} = \frac{(\sum w)^2}{\sum w^2}$$

The middle term is the error all polls of a state share, so more polls can't remove it.

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4. NATIONAL ENVIRONMENT

$$\text{env} = 0.75 \times \text{GB} + 0.25 \times 0.35 \times (-\text{net approval}), \qquad \text{signal} = \frac{n_{\text{fresh}} \cdot \text{VoteHub} + 5 \cdot \text{Kalshi}}{n_{\text{fresh}} + 5}$$

GB = generic ballot. Each VoteHub poll counts 0.5^(age / 14) toward n_fresh. This run: env = D+8.0.

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5. PRIOR (FUNDAMENTALS)

$$\text{prior} = 2 \times \text{PVI} + \text{env} \pm 3.78\,[\text{incumbent}] + 0.30 \times c, \qquad SD_{\text{prior}} = 9.32$$

Appointed incumbents get 1 pt. c = candidate effect: the nominees' past margin minus a generic candidate's (capped at +/- 25). Independent-led races: SD x 1.3.

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6. KALSHI

$$P' = \text{logistic}\,(1.15 \cdot \text{logit}\,P), \qquad \text{margin}_K = \Phi^{-1}(P') \cdot \max\,(4.42,\ SD_{F+P}), \qquad SD_K = 8$$

Used only when 6% <= P <= 94%.

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7. EACH RACE

$$\text{margin} = \frac{\pi_F \cdot \text{prior} + \pi_P \cdot \bar{m} + \pi_K \cdot \text{margin}_K}{\pi_F + \pi_P + \pi_K}, \quad \pi = \frac{1}{SD^2}, \quad SD^2 = \frac{1}{\Pi} + \frac{2(0.3)}{\Pi^2}\left(\sqrt{\pi_F\pi_P} + \sqrt{\pi_F\pi_K} + \sqrt{\pi_P\pi_K}\right) + 0.76^2 \cdot \text{days left}$$

Pi = the sum of the three precisions. The second term widens the range because the three sources tend to miss together.

$$\tilde{m}_i = \text{margin}_i + N + R_{g(i)} + L\,f_i + e_i, \qquad \text{D seats} = 34 + \sum_i \mathbf{1}[\tilde{m}_i > 0], \qquad P(D \geq 51) = \text{share of simulations}$$

N = national miss (SD 2.52, t with df 8). R = regional miss (SD 1.51). L = lean factor (SD 1), f_i = PVI / 10. e = the rest of the race's SD (t with df 10.9). An independent's win counts for Democrats with probability: Nebraska 0.8.